解答　15　直線上の点　練習問題

1　2点間の距離

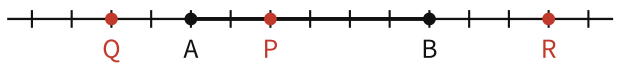
(1)	$AB = \lvert 1-5 \rvert = \lvert -4 \rvert = 4$	4
(2)	$AB = \lvert 3-(-4) \rvert = \lvert 7 \rvert = 7$	7
(3)	$AB = \lvert -6-(-3) \rvert = \lvert -3 \rvert = 3$	3
(4)	$OA = \lvert 10-0 \rvert = \lvert 10 \rvert = 10$	10
(5)	$AB = \lvert 7-5 \rvert = \lvert 2 \rvert = 2$	2
(6)	$AB = \lvert 1-(-7) \rvert = \lvert 8 \rvert = 8$	8
(7)	$AB = \lvert -2-(-9) \rvert = \lvert 7 \rvert = 7$	7
(8)	$OA = \lvert 6-0 \rvert = \lvert 6 \rvert = 6$	6
(9)	$AB = \lvert 5-3 \rvert = \lvert 2 \rvert = 2$	2
(10)	$AB = \lvert 3-(-8) \rvert = \lvert 11 \rvert = 11$	11
(11)	$AB = \lvert -1-(-5) \rvert = \lvert 4 \rvert = 4$	4
(12)	$OA = \lvert -3-0 \rvert = \lvert -3 \rvert = 3$	3
(13)	$AB = \lvert 2-5 \rvert = \lvert -3 \rvert = 3$	3
(14)	$AB = \lvert 3-(-1) \rvert = \lvert 4 \rvert = 4$	4
(15)	$AB = \lvert -1-(-3) \rvert = \lvert 2 \rvert = 2$	2
(16)	$OA = \lvert -11-0 \rvert = \lvert -11 \rvert = 11$	11

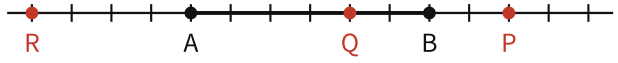
2　内分・外分の比

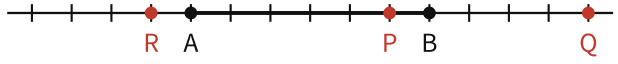
①	$A(-5),\ B(-1),\ C(0)$	
(1)	B は AC を	$4:1$ **に内分**
(2)	C は AB を	$5:1$ **に外分**
(3)	A は BC を	$4:5$ **に外分**
②	$A(0),\ B(5),\ C(6)$	
(1)	B は AC を	$5:1$ **に内分**
(2)	C は AB を	$6:1$ **に外分**
(3)	A は BC を	$5:6$ **に外分**
③	$A(-1),\ B(5),\ C(1)$	
(1)	B は AC を	$3:2$ **に外分**
(2)	C は AB を	$1:2$ **に内分**
(3)	A は BC を	$3:1$ **に外分**
④	$A(3),\ B(-3),\ C(5)$	
(1)	B は AC を	$3:4$ **に外分**
(2)	C は AB を	$1:4$ **に外分**
(3)	A は BC を	$3:1$ **に内分**

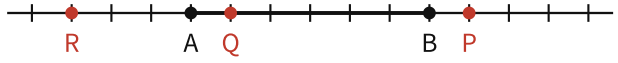
解答　15　直線上の点　練習問題

3　図示

① 

② 

③ 

④ 

4　内分点・外分点の座標

① $A(4),\ B(14)$

(1)	$\dfrac{4\cdot 4+1\cdot 14}{1+4}=\dfrac{30}{5}=6$	$P(6)$
(2)	$\dfrac{2\cdot 4+3\cdot 14}{3+2}=\dfrac{50}{5}=10$	$Q(10)$
(3)	$\dfrac{-1\cdot 4+3\cdot 14}{3-1}=\dfrac{38}{2}=19$	$R(19)$
(4)	$\dfrac{4+14}{2}=\dfrac{18}{2}=9$	$M(9)$

② $A(-4),\ B(12)$

(5)	$\dfrac{3\cdot(-4)+5\cdot 12}{5+3}=\dfrac{48}{8}=6$	$P(6)$
(6)	$\dfrac{5\cdot(-4)+3\cdot 12}{3+5}=\dfrac{16}{8}=2$	$Q(2)$
(7)	$\dfrac{-5\cdot(-4)+3\cdot 12}{3-5}=\dfrac{56}{-2}=-28$	$R(-28)$
(8)	$\dfrac{-4+12}{2}=\dfrac{8}{2}=4$	$M(4)$

③ $A(1),\ B(11)$

(9)	$\dfrac{4\cdot 1+1\cdot 11}{1+4}=\dfrac{15}{5}=3$	$P(3)$
(10)	$\dfrac{3\cdot 1+2\cdot 11}{2+3}=\dfrac{25}{5}=5$	$Q(5)$
(11)	$\dfrac{-3\cdot 1+1\cdot 11}{1-3}=\dfrac{8}{-2}=-4$	$R(-4)$
(12)	$\dfrac{1+11}{2}=\dfrac{12}{2}=6$	$M(6)$

④ $A(1),\ B(13)$

(13)	$\dfrac{1\cdot 1+2\cdot 13}{2+1}=\dfrac{27}{3}=9$	$P(9)$
(14)	$\dfrac{1\cdot 1+3\cdot 13}{3+1}=\dfrac{40}{4}=10$	$Q(10)$
(15)	$\dfrac{-3\cdot 1+1\cdot 13}{1-3}=\dfrac{10}{-2}=-5$	$R(-5)$
(16)	$\dfrac{1+13}{2}=\dfrac{14}{2}=7$	$M(7)$